Lecture 8

The harmonic oscillator and coherent states

Objectives. *The Hermite functions form a complete eigenbasis for the harmonic oscillator. Its energy is the only self-adjoint operator that acts by the oscillator rule on rapidly decreasing functions. Its evolution is periodic, and a quarter period is a Fourier transform, up to a change of scale and a phase. Coherent states follow classical orbits without spreading.*

- Construct Hermite functions with the ladder operators.
- Compute the Poisson distribution of level numbers in a coherent state.
- Explain why nonorthogonal coherent states cannot be distinguished with certainty.

Lecture 8, first part

The oscillator's levels and the Hermite basis

8.1 Ladder operators

The free energy H_0 on the line has no eigenvalue (Theorem 4.6), so a free particle has no energy level, and its packets spread at late times (Proposition 7.12). A spring gives levels on the whole line. In this lecture we find them, and we follow the evolution that they give.

A particle of mass m bound to the origin by a spring, with the classical angular frequency ν, has the classical energy $p^2/2m + m\nu^2x^2/2$. On a rapidly decreasing function $\psi \in \mathcal{S}(\mathbb{R})$, the corresponding rule is

$$-\frac{\hbar^2}{2m}\,\psi'' + \frac{m\nu^2}{2}\,x^2\psi, \tag{8.1}$$

which is $P^2\psi/2m + m\nu^2Q^2\psi/2$ by Theorem 4.6. The rule is not yet an operator with a domain. § 8.3 finds the self-adjoint operator. The oscillator has a natural length, the *oscillator length*

$$s = \sqrt{\frac{\hbar}{m\nu}},$$

at which the two energies $\hbar^2/ms^2$ and $m\nu^2s^2$ both equal $\hbar\nu$. It is not the width s of Lecture 4. The state of lowest energy, found below, is a Gaussian packet with the spread $\Delta Q = s/\sqrt{2}$.

Definition 8.1 (Ladder operators)**.** On $\mathcal{S}(\mathbb{R})$, the *lowering operator* a and the *raising operator* $a^\dagger$ are

$$a = \left(\frac{m\nu}{2\hbar}\right)^{1/2}\left(Q + \frac{\mathrm{i}P}{m\nu}\right) \quad \text{and} \quad a^\dagger = \left(\frac{m\nu}{2\hbar}\right)^{1/2}\left(Q - \frac{\mathrm{i}P}{m\nu}\right).$$

Since $(m\nu/2\hbar)^{1/2} = 1/\sqrt{2}\,s$ and $\hbar/m\nu = s^2$, the ladder operators act by

$$a\psi = \frac{1}{\sqrt{2}}\Big(\frac{x\psi}{s} + s\psi'\Big) \quad \text{and} \quad a^\dagger\psi = \frac{1}{\sqrt{2}}\Big(\frac{x\psi}{s} - s\psi'\Big).$$

On this test space, $a^\dagger$ denotes the raising rule just defined. The next identity says that it is a restriction of the Hilbert-space adjoint of a on $\mathcal{S}(\mathbb{R})$; it does not identify their full domains.

Proposition 8.2 (The ladder algebra)**.** *The operators a and $a^\dagger$ map $\mathcal{S}(\mathbb{R})$ into itself. For all $\varphi, \psi \in \mathcal{S}(\mathbb{R})$,*

$$\langle \varphi, a\psi\rangle = \left\langle a^\dagger\varphi, \psi\right\rangle, \qquad aa^\dagger\psi - a^\dagger a\psi = \psi,$$

and the rule (8.1) equals $\hbar\nu(a^\dagger a + \frac{1}{2})\psi$.

Proof. By Proposition 4.7, Q and P map $\mathcal{S}(\mathbb{R})$ into itself, so a and $a^\dagger$ do. Since Q and P are symmetric, the coefficient $(m\nu/2\hbar)^{1/2}$ is real, and the inner product is antilinear in its first slot,

$$\langle \varphi, (Q + \mathrm{i}P/m\nu)\psi\rangle = \langle (Q - \mathrm{i}P/m\nu)\varphi, \psi\rangle\,.$$

Next, let $\beta = 1/m\nu$. Expanding the products, on $\mathcal{S}(\mathbb{R})$,

$$(Q + \mathrm{i}\beta P)(Q - \mathrm{i}\beta P) - (Q - \mathrm{i}\beta P)(Q + \mathrm{i}\beta P) = -2\mathrm{i}\beta(QP - PQ) = 2\beta\hbar,$$

by Proposition 4.7, and $(m\nu/2\hbar)\,2\beta\hbar = 1$. In the same way, $(Q - \mathrm{i}\beta P)(Q + \mathrm{i}\beta P) = Q^2 + \beta^2P^2 - \beta\hbar$, so

$$\hbar\nu\, a^\dagger a = \frac{m\nu^2}{2}\Big(Q^2 + \frac{P^2}{m^2\nu^2}\Big) - \frac{\hbar\nu}{2} = \frac{P^2}{2m} + \frac{m\nu^2Q^2}{2} - \frac{\hbar\nu}{2}.$$

∎

The equation $a\psi = 0$ on $\mathcal{S}(\mathbb{R})$ reads $\psi' = -x\psi/s^2$, that is, $(e^{x^2/2s^2}\psi)' = 0$. So ψ is a multiple of $e^{-x^2/2s^2}$, and the unit vector with a positive multiple is

$$h_0(x) = (\pi s^2)^{-1/4} e^{-x^2/2s^2}.$$

This is the packet (4.3) with $q = p_0 = 0$ and the width $s/\sqrt{2}$, so its position variance is $s^2/2$. So the oscillator length s is $\sqrt{2}$ times the width of this packet.

Proposition 8.3 (The Hermite functions)**.** *Let $h_n = (n!)^{-1/2}(a^\dagger)^n h_0$ for $n \geq 0$, and let $g(x) = e^{-x^2/2s^2}$.*

1. *Each h_n is $r_n g$ for a polynomial r_n of degree n with a positive leading coefficient, and $h_n(-x) = (-1)^n h_n(x)$. So $h_n \in \mathcal{S}(\mathbb{R})$.*
2. *$a^\dagger h_n = \sqrt{n+1}\,h_{n+1}$, $ah_0 = 0$, and $ah_n = \sqrt{n}\,h_{n-1}$ for $n \geq 1$. So $a^\dagger a h_n = nh_n$, and the rule (8.1) maps h_n to $\hbar\nu(n + \frac{1}{2})h_n$.*
3. *The h_n are orthonormal.*

Proof.

1. For a polynomial r, $a^\dagger(rg) = (2xr/s - sr')g/\sqrt{2}$. If r has degree n, a positive leading coefficient c and the parity of n, then $2xr/s - sr'$ has degree $n+1$, the leading coefficient $2c/s$ and the parity of $n+1$. Induction from $r_0 = (\pi s^2)^{-1/4}$ gives the claim, and $r_n g \in \mathcal{S}(\mathbb{R})$ by Definition 4.1.
2. The first relation is the definition. By Proposition 8.2, $aa^\dagger = a^\dagger a + 1$ on $\mathcal{S}(\mathbb{R})$. So $ah_1 = aa^\dagger h_0 = h_0$, and if $ah_n = \sqrt{n}\,h_{n-1}$, then $a^\dagger a h_n = nh_n$ and

$$ah_{n+1} = \frac{aa^\dagger h_n}{\sqrt{n+1}} = \frac{(n+1)h_n}{\sqrt{n+1}} = \sqrt{n+1}\,h_n.$$

The last claim follows from Proposition 8.2.

3. The h_n are real. For $n \geq 1$, the adjoint identity of Proposition 8.2 gives $\langle h_0, h_n\rangle = \langle a h_0, h_{n-1}\rangle / \sqrt{n} = 0$, and for $m, n \geq 1$

$$\langle h_m, h_n\rangle = \frac{1}{\sqrt{n}} \left\langle h_m, a^\dagger h_{n-1}\right\rangle = \frac{1}{\sqrt{n}} \langle a h_m, h_{n-1}\rangle = \sqrt{\frac{m}{n}}\, \langle h_{m-1}, h_{n-1}\rangle .$$

Applying this $\min(m,n)$ times reaches $\left\langle h_0, h_{|m-n|}\right\rangle$, which is 0 unless $m = n$, and then 1. ∎

For example, $h_1 = \sqrt{2}\,(x/s)h_0$. In general $h_n = (2^n n!)^{-1/2} H_n(x/s) h_0$, where H_n is the Hermite polynomial with $H_0 = 1$ and $H_{n+1}(y) = 2yH_n(y) - H_n'(y)$.

The ladder algebra built the h_n. It cannot tell whether they are all the oscillator's states. A vector orthogonal to every h_n need not lie in $\mathcal{S}(\mathbb{R})$, where a acts, so the algebra never reaches it (Hall, pp. 231–235). The next section proves that no such nonzero vector exists.

Self-check 8.1

1. Compute h_2, and check that $\langle h_0, h_2\rangle = 0$.

8.2 The Hermite functions are complete

Linear Algebra, Ex. 13.35, states that the Hermite functions form a basis of $L^2(\mathbb{R})$. Here is a proof from the standing fact S1.

Theorem 8.4 (The Hermite functions are complete)**.** *The functions h_n, $n \geq 0$, form an orthonormal basis of $L^2(\mathbb{R})$.*

Proof. Let $\varphi \in L^2(\mathbb{R})$ be orthogonal to every h_n. We show that $\varphi = 0$. Then, by Lemma 6.5 applied to the closed span of the h_n, that span is $L^2(\mathbb{R})$. We first show that φ is orthogonal to every polynomial times g, and then to every plane wave times g. This says that the transform of $g\overline{\varphi}$ is 0, and the standing fact S1 then gives $\varphi = 0$.

1. *Polynomials.* By Proposition 8.3, $h_0, \dots, h_N$ and $g, xg, \dots, x^N g$ span the same space, since r_n has degree n. So $\langle \varphi, x^n g\rangle = 0$ for every n.
2. *A Taylor bound.* For real θ and $N \geq 0$, let $\delta_N(\theta) = e^{-\mathrm{i}\theta} - \sum_{n=0}^{N} (-\mathrm{i}\theta)^n/n!$, and $\delta_{-1}(\theta) = e^{-\mathrm{i}\theta}$. Then $\delta_N(0) = 0$ and $\delta_N' = -\mathrm{i}\delta_{N-1}$, so $\delta_N(\theta) = -\mathrm{i}\int_0^\theta \delta_{N-1}(u)\,\mathrm{d}u$, and by induction on N

$$|\delta_N(\theta)| \leq \frac{|\theta|^{N+1}}{(N+1)!}. \tag{8.2}$$

3. *Moments.* Let $I_n = \int x^{2n} e^{-x^2/s^2}\,\mathrm{d}x = \|x^n g\|^2$. Then $I_0 = s\sqrt{\pi}$, and integration by parts of $x^{2n-1}\cdot x e^{-x^2/s^2}$ gives $I_n = \frac{1}{2}(2n-1)s^2 I_{n-1}$. Since $\frac{1}{2}(2k-1) \leq k$, $I_n \leq n!\, s^{2n} I_0$.
4. *Plane waves.* Fix a real p, and let $S_N = \sum_{n=0}^{N} (-\mathrm{i}p/\hbar)^n x^n g/n!$. By (8.2) with $\theta = px/\hbar$, and by the moments,

$$\left\| e^{-\mathrm{i}px/\hbar} g - S_N \right\|^2 \leq \left(\frac{|p|}{\hbar}\right)^{2N+2} \frac{I_{N+1}}{(N+1)!^2} \leq I_0 \frac{(|p|\,s/\hbar)^{2N+2}}{(N+1)!},$$

which tends to 0. By the first step, $\langle \varphi, S_N\rangle = 0$, so

$$\left\langle \varphi, e^{-\mathrm{i}px/\hbar} g\right\rangle = \int_{-\infty}^{\infty} e^{-\mathrm{i}px/\hbar} g(x)\overline{\varphi(x)}\,\mathrm{d}x = 0 \quad \text{for every real } p. \tag{8.3}$$

5. *The transform.* Let $u = g\overline{\varphi}$. Since $0 < g \le 1$, u is square-integrable. By Cauchy–Schwarz, $\int_{|x|>A} |u|\,\mathrm{d}x \le \left\|1_{|x|>A}\, g\right\| \, \|\varphi\|$, which is finite for $A = 0$ and tends to 0 as $A \to \infty$. By Hall, Thm A.19, which states S1 for $L^2(\mathbb{R})$, the transform Fu is the limit in norm of

$$u_A(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-A}^{A} e^{-\mathrm{i}px/\hbar} u(x)\,\mathrm{d}x$$

as $A \to \infty$. By (8.3), the integral over the whole line vanishes, so $u_A(p)$ is minus the integral over $|x| > A$, and $|u_A(p)| \le (2\pi\hbar)^{-1/2} \int_{|x|>A} |u|\,\mathrm{d}x$ for every p, so $u_A \to 0$ uniformly. On a bounded interval J, therefore, $\int_J |u_A|^2\,\mathrm{d}p \to 0$, while $\int_J |Fu - u_A|^2\,\mathrm{d}p \le \|Fu - u_A\|^2 \to 0$. So Fu vanishes on every bounded interval, and $Fu = 0$. By S1, $u = 0$, and since $g > 0$, $\varphi = 0$. ∎

So every $\psi \in L^2(\mathbb{R})$ is $\sum_n \langle h_n, \psi\rangle\, h_n$, and $\|\psi\|^2 = \sum_n |\langle h_n, \psi\rangle|^2$.

Self-check 8.2

1. Let $\varphi \in L^2(\mathbb{R})$ be orthogonal to every h_n with n even. Show that φ is odd.

8.3 The oscillator's levels

The box has only discrete spectrum, and the free particle on the line has none. The difference shows in the resolvent. Recall that a bounded operator is *compact* if it is the limit in norm of operators of finite rank (*Linear Algebra*, Def. 14.19). A compact resolvent records the discreteness of the levels in a single bounded operator. For an operator built from an eigenbasis, we prove that discrete levels, each of finite multiplicity and with no finite limit point, give a compact resolvent, and Theorem 8.7 applies this to the oscillator. The converse holds for every self-adjoint operator, and we do not use it. Lecture 10 returns to compact operators, in Weyl's theorem on the essential spectrum.

Proposition 8.5 (Compact resolvent). *Let B be the operator made from an orthonormal basis (e_k) and real numbers λ_k, as in Proposition 5.5, and suppose that each bounded interval contains λ_k for only finitely many k. Then $\sigma(B) = \sigma_{\mathrm{d}}(B)$ is the set of the λ_k, and $(B - z)^{-1}$ is compact for every z in the resolvent set.*

Proof. The set of the λ_k has no limit point, so it is closed, and by Proposition 6.8 it is $\sigma(B)$. If $B\psi = \lambda_k\psi$, then $\lambda_l c_l = \lambda_k c_l$ for every l, so the eigenspace for λ_k is spanned by the finitely many e_l with $\lambda_l = \lambda_k$. Since a bounded interval holds only finitely many λ_l, a small interval around λ_k contains no other point of $\sigma(B)$. So $\sigma(B) = \sigma_{\mathrm{d}}(B)$. For z in the resolvent set, let R_N keep the terms of $(B - z)^{-1}\psi = \sum_k c_k(\lambda_k - z)^{-1} e_k$ with $|\lambda_k| \le N$. It has finite rank, since only finitely many k have $|\lambda_k| \le N$, and for $N > |z|$ Parseval's identity gives

$$\left\|(B - z)^{-1} - R_N\right\| \le \sup_{k:\,|\lambda_k|>N} \frac{1}{|\lambda_k - z|} \le \frac{1}{N - |z|},$$

which tends to 0. So $(B - z)^{-1}$ is compact. ∎

The box Hamiltonian, the momentum on the circle and each P_θ (Theorem 3.10) have compact resolvents. A particle between walls has discrete levels, and a free particle has none. A spring also gives discrete levels.

Definition 8.6 (The oscillator's Hamiltonian)**.** The *Hamiltonian* H of the harmonic oscillator is the operator made, as in Proposition 5.5, from the basis (h_n) and the numbers $\hbar\nu(n+\frac{1}{2})$: with $c_n = \langle h_n, \psi\rangle$,

$$D(H) = \Big\{\psi : \sum_n (n+\tfrac{1}{2})^2 |c_n|^2 < \infty\Big\}, \qquad H\psi = \sum_n \hbar\nu(n+\tfrac{1}{2}) c_n h_n.$$

Theorem 8.7 (The oscillator's levels)**.** *The operator H is self-adjoint. Every $\psi \in \mathcal{S}(\mathbb{R})$ lies in $D(H)$, and $H\psi$ is* (8.1)*. Moreover, H is the only self-adjoint operator that acts on $\mathcal{S}(\mathbb{R})$ by* (8.1)*. Its spectrum is*

$$\sigma(H) = \sigma_{\mathrm{d}}(H) = \big\{\hbar\nu(n+\tfrac{1}{2}) : \ n = 0, 1, 2, \dots\big\},$$

each level is simple, and $(H-z)^{-1}$ is compact for every z off the spectrum.

Proof. By Theorem 8.4, (h_n) is an orthonormal basis, so H is self-adjoint by Proposition 5.5. Let $\psi \in \mathcal{S}(\mathbb{R})$, and let $\chi \in \mathcal{S}(\mathbb{R})$ be (8.1). By Propositions 8.2 and 8.3, and since $\hbar\nu(n+\frac{1}{2})$ is real,

$$\langle h_n, \chi\rangle = \hbar\nu \left\langle h_n, (a^\dagger a + \tfrac{1}{2})\psi\right\rangle = \hbar\nu \left\langle (a^\dagger a + \tfrac{1}{2}) h_n, \psi\right\rangle = \hbar\nu(n+\tfrac{1}{2}) c_n.$$

By Parseval's identity, $\sum_n \hbar^2\nu^2 (n+\frac{1}{2})^2 |c_n|^2 = \|\chi\|^2$ is finite, so $\psi \in D(H)$, and $H\psi = \sum_n \langle h_n, \chi\rangle h_n = \chi$. Next, let A be self-adjoint and act on $\mathcal{S}(\mathbb{R})$ by (8.1). By Proposition 8.3, $Ah_n = \hbar\nu(n+\frac{1}{2})h_n$ for every n, so $A = H$ by Proposition 5.5. Finally, a bounded interval contains finitely many of the numbers $\hbar\nu(n+\frac{1}{2})$, so Proposition 8.5 gives the spectrum and the compact resolvent. If $H\psi = \hbar\nu(n+\frac{1}{2})\psi$, then $\hbar\nu(m-n)c_m = 0$ for every m, so ψ is a multiple of h_n. So each level is simple. ∎

In the box, the rule $-\mathrm{i}\hbar\psi'$ belongs to many self-adjoint operators, the P_θ of Theorem 3.10. The oscillator's rule belongs to one, so its energy needs no choice of boundary condition.

Figure 8.1 draws the first five levels with their Hermite functions, inside the parabola of the classical potential energy $m\nu^2x^2/2$. A classical particle with the energy $\hbar\nu(n+\frac{1}{2})$ turns back at $x = \sqrt{2n+1}\,s$ and at $x = -\sqrt{2n+1}\,s$, where the parabola meets the level. In the figure, h_n has n zeros and the parity of n. It does not end at the turning points, since a polynomial times a Gaussian vanishes at only finitely many points. So a position reading in the state h_n falls beyond the turning points with positive probability.

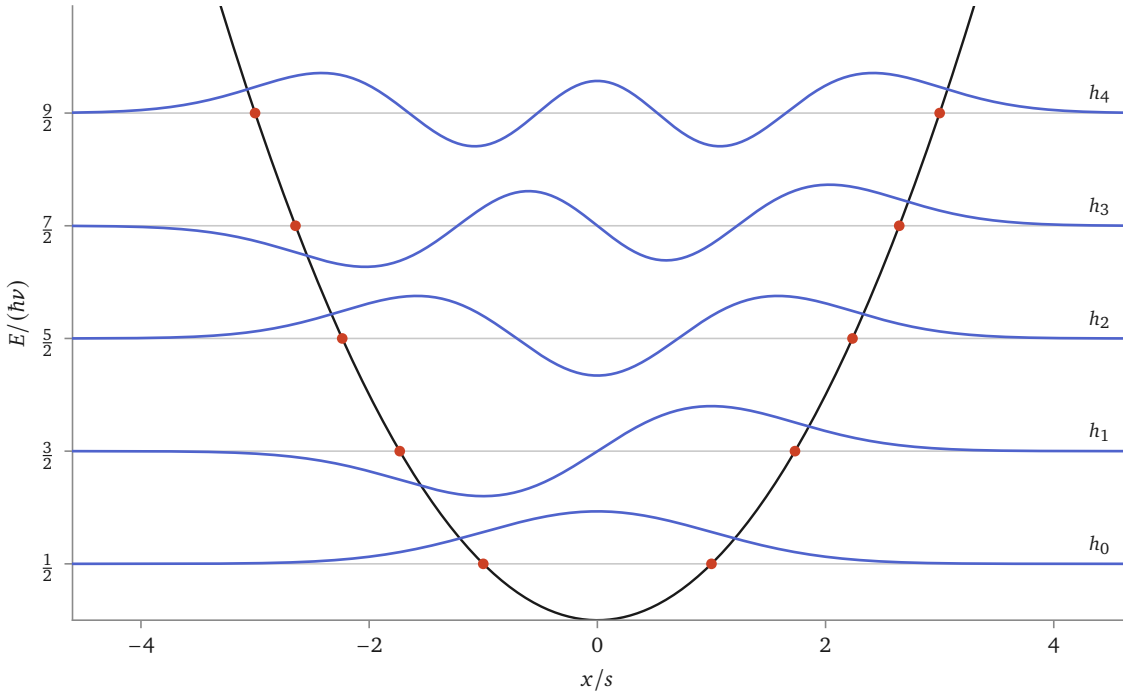


Figure 8.1: The parabola is the classical potential energy, in units of $\hbar\nu$. Each horizontal line is a level $n + \frac{1}{2}$, for $n = 0, \dots, 4$, and the curve on it is the signed function h_n, drawn about its level, with one vertical scale for all five. The height of a curve above its level is an amplitude in arbitrary units, not an energy and not a density. The dots mark the classical turning points.

The lowest level is $\hbar\nu/2$, not 0. The oscillator cannot rest at the bottom of its well, since a state with small ΔQ has a large ΔP (self-check).

Self-check 8.3

1. Let $\psi \in \mathcal{S}(\mathbb{R})$ be a unit vector. Show from Theorem 4.9 that $\langle \psi, H\psi \rangle \geq \hbar\nu/2$.

Lecture 8, second part

The oscillator's evolution and coherent states

8.4 A quarter period is a Fourier transform

The first part found the oscillator's Hamiltonian H. Its eigenvectors, the Hermite functions, form an orthonormal basis, and its levels are $\hbar\nu(n + \frac{1}{2})$. In this part we follow the evolution that H generates. Every state returns after one period, a quarter period gives a Fourier transform, up to a common phase and a change of scale, and the coherent states follow classical orbits without spreading.

This part and Lecture 9 use the integral G of Example 4.12 in the following form.

Lemma 8.8 (A Gaussian integral). *For every $b > 0$ and every real k,*

$$\int_{-\infty}^{\infty} e^{-ikx} e^{-x^2/b^2} \, dx = b\sqrt{\pi}\, e^{-b^2k^2/4}.$$

Proof. The first step of Example 4.12 proves

$$\int_{-\infty}^{\infty} e^{-\mathrm{i}ky} e^{-y^2/4s^2}\,\mathrm{d}y = 2s\sqrt{\pi}\,e^{-s^2k^2}$$

for every width $s > 0$ and every real k. Put $s = b/2$. Then $4s^2 = b^2$, $2s = b$ and $s^2k^2 = b^2k^2/4$, which gives the claim. ∎

By Theorem 7.4, the operators

$$U_t\psi = e^{-\mathrm{i}tH/\hbar}\psi = \sum_n e^{-\mathrm{i}\nu(n+\frac{1}{2})t} c_n h_n$$

form a strongly continuous unitary group with the generator $H/\hbar$. After one period $2\pi/\nu$, every phase is $e^{-\mathrm{i}\pi(2n+1)} = -1$, so $U_{2\pi/\nu} = -\mathbb{1}$. Every state returns to itself, up to the common sign -1, which changes no reading.

After half a period the phases are $e^{-\mathrm{i}\pi(n+\frac{1}{2})} = -\mathrm{i}(-1)^n$. The parity $(\Pi\psi)(x) = \psi(-x)$ is unitary, and $\Pi h_n = (-1)^n h_n$ by Proposition 8.3. Two bounded operators that agree on an orthonormal basis agree on its finite combinations, and so, by continuity, everywhere. So $U_{\pi/\nu} = -\mathrm{i}\Pi$, that is,

$$(U_{\pi/\nu}\psi)(x) = -\mathrm{i}\,\psi(-x).$$

Every state is mirrored, as the box's states are at half the revival time (Proposition 2.4). After a quarter of a period, the position wavefunction is the initial momentum wavefunction, up to a common phase and a change of scale. Let D be the unitary

$$(D\psi)(p) = (m\nu)^{-1/2}\,\psi(p/m\nu),$$

which reads a function of x as a function of $p = m\nu x$. Its inverse is $(D^{-1}\varphi)(x) = (m\nu)^{1/2}\varphi(m\nu x)$.

Proposition 8.9 (A quarter period). *For every n, $Fh_n = (-\mathrm{i})^n Dh_n$. For every $\psi \in L^2(\mathbb{R})$,*

$$U_{\pi/2\nu} = e^{-\mathrm{i}\pi/4}D^{-1}F, \quad \textit{that is,} \quad (U_{\pi/2\nu}\psi)(x) = e^{-\mathrm{i}\pi/4}(m\nu)^{1/2}\tilde{\psi}(m\nu x).$$

Proof. First, $Fh_0 = Dh_0$. By Lemma 8.8 with $b = \sqrt{2}\,s$ and $k = p/\hbar$,

$$Fh_0(p) = \left(\frac{s^2}{\pi\hbar^2}\right)^{1/4} e^{-s^2p^2/2\hbar^2}.$$

Since $m^2\nu^2s^2 = \hbar^2/s^2$, this is $Dh_0(p)$. Next, suppose that $Fh_n = (-\mathrm{i})^n Dh_n$, which lies in $\mathcal{S}(\mathbb{R})$. By Lemma 7.11 and Proposition 4.3, the transforms of Qh_n and Ph_n are $\mathrm{i}\hbar(Fh_n)'$ and pFh_n, so

$$F(a^\dagger h_n) = \frac{1}{\sqrt{2}\,s}\left(\mathrm{i}\hbar(Fh_n)' - \frac{\mathrm{i}p}{m\nu}Fh_n\right) = \frac{-\mathrm{i}}{\sqrt{2}\,s}\left(\frac{p}{m\nu} - \hbar\frac{\mathrm{d}}{\mathrm{d}p}\right)Fh_n.$$

For every $\varphi \in \mathcal{S}(\mathbb{R})$, the chain rule gives $\hbar(D\varphi)' = s^2 D(\varphi')$, since $\hbar/m\nu = s^2$, and $(p/m\nu)D\varphi = D(x\varphi)$. So

$$\frac{1}{\sqrt{2}\,s}\left(\frac{p}{m\nu} - \hbar\frac{\mathrm{d}}{\mathrm{d}p}\right)D\varphi = D\,\frac{1}{\sqrt{2}}\left(\frac{x\varphi}{s} - s\varphi'\right) = Da^\dagger\varphi.$$

With $\varphi = h_n$ and $Fh_n = (-\mathrm{i})^n Dh_n$, this gives $F(a^\dagger h_n) = (-\mathrm{i})^{n+1} Da^\dagger h_n$, and dividing by $\sqrt{n+1}$ gives $Fh_{n+1} = (-\mathrm{i})^{n+1} Dh_{n+1}$. Finally, $e^{-\mathrm{i}\nu(n+\frac{1}{2})\pi/2\nu} = e^{-\mathrm{i}\pi/4}(-\mathrm{i})^n$, so the bounded operators $U_{\pi/2\nu}$ and $e^{-\mathrm{i}\pi/4}D^{-1}F$ agree on every h_n, and so everywhere. ∎

In words, after a quarter period the position density is $m\nu\left|\tilde{\psi}(m\nu x)\right|^2$. The probability that the position then lies in an interval I is the probability that the initial momentum lies in $m\nu I$. A classical oscillator does the same. It moves as $x\cos\nu t + (p_0/m\nu)\sin\nu t$, and after a quarter period it is at $p_0/m\nu$. In units with $s = \hbar = 1$, $m\nu = 1$, so $D = \mathbb{1}$ and $Fh_n = (-\mathrm{i})^n h_n$. So the Hermite functions are eigenvectors of the Fourier transform.

Figure 8.2 shows this for the packet (4.3) with $q = p_0 = 0$ and the width $0.4\,s/\sqrt{2}$, which is narrower than h_0. By Theorem 4.9, its momentum spread is 2.5 times that of h_0, so a quarter period later its position spread is $2.5\,s/\sqrt{2}$. After half a period its density is the initial one again, since $U_{\pi/\nu} = -\mathrm{i}\Pi$ and the packet is even.

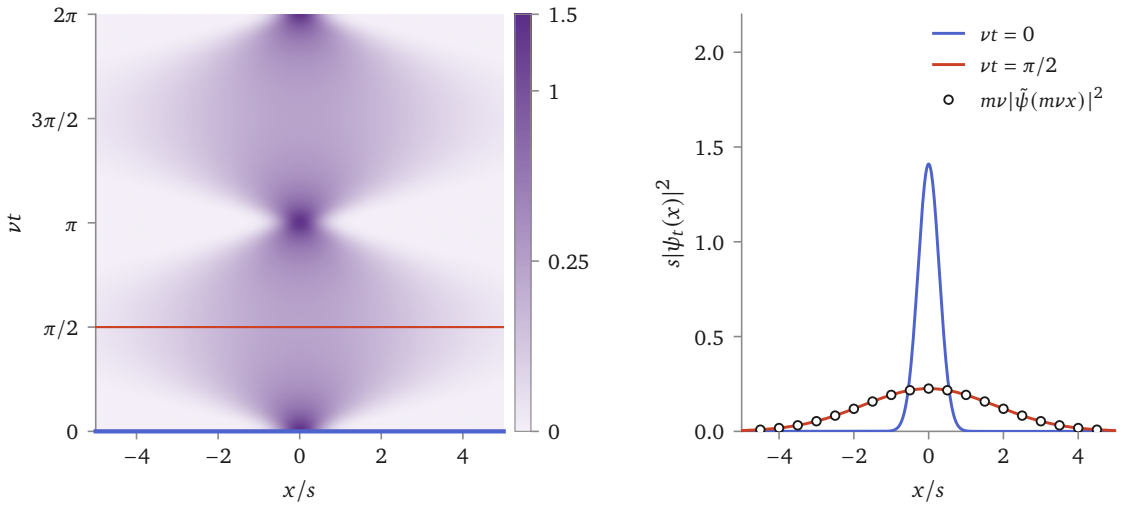


Figure 8.2: The packet (4.3) with $q = p_0 = 0$ and the width $0.4\,s/\sqrt{2}$, over one period, in units with $s = \hbar = 1$, so that $m\nu = 1$. Left: the position density $|\psi_t(x)|^2$ of $\psi_t = U_t\psi$, with x/s across and νt upwards, darker where it is larger. The key gives $s\,|\psi_t(x)|^2$ on a square-root scale. The density is narrow at $\nu t = 0$, π and 2π, and wide at $\nu t = \pi/2$ and $3\pi/2$. Right: the densities at $\nu t = 0$ and $\nu t = \pi/2$, the times of the two coloured lines on the left. The dots are $m\nu\left|\tilde{\psi}(m\nu x)\right|^2$, computed from the initial packet. They lie on the curve at $\nu t = \pi/2$, as Proposition 8.9 says.

Self-check 8.4

1. The transform $\tilde{\psi}$ of a state vanishes for $p < 0$. What is the probability that the position is negative a quarter period later?

8.5 Coherent states

An eigenvector of a gives a packet whose means follow a classical orbit.

Theorem 8.10 (Coherent states). *For a complex number α, let*

$$\psi_\alpha = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,h_n,$$

and write $\alpha = (q/s + \mathrm{i}sp_0/\hbar)/\sqrt{2}$ with q and p_0 real.

1. $\psi_\alpha(x) = e^{-\mathrm{i}qp_0/2\hbar} e^{\mathrm{i}p_0 x/\hbar} h_0(x-q)$, *so* $\psi_\alpha \in \mathcal{S}(\mathbb{R})$, *and* $a\psi_\alpha = \alpha\psi_\alpha$.
2. ψ_α *is the packet* (4.3) *with the width* $s/\sqrt{2}$*:* $\langle Q\rangle = q$, $\langle P\rangle = p_0$, $\Delta Q = s/\sqrt{2}$ *and* $\Delta P = \hbar/\sqrt{2}\,s$, *so* $\Delta Q\,\Delta P = \hbar/2$.
3. $|\langle h_n, \psi_\alpha\rangle|^2 = e^{-|\alpha|^2}\,|\alpha|^{2n}/n!$, *a Poisson distribution of* n *with the mean* $|\alpha|^2$.
4. $U_t\psi_\alpha = e^{-\mathrm{i}\nu t/2}\psi_{\alpha e^{-\mathrm{i}\nu t}}$ *for every real* t.

Proof.

1. Let φ be the function on the right. It is a Gaussian packet, so it lies in $\mathcal{S}(\mathbb{R})$, and $s\varphi' = (-(x-q)/s + \mathrm{i}sp_0/\hbar)\varphi$. So $a\varphi = (x\varphi/s + s\varphi')/\sqrt{2} = \alpha\varphi$. By the adjoint identity of Proposition 8.2, used n times,

$$\langle h_n, \varphi\rangle = \frac{1}{\sqrt{n!}}\left\langle (a^\dagger)^n h_0, \varphi\right\rangle = \frac{1}{\sqrt{n!}}\,\langle h_0, a^n\varphi\rangle = \frac{\alpha^n}{\sqrt{n!}}\,\langle h_0, \varphi\rangle\,.$$

   To compute $\langle h_0, \varphi\rangle$, write $x^2 + (x-q)^2 = 2(x-q/2)^2 + q^2/2$ and substitute $x = y + q/2$. The phase $e^{\mathrm{i}p_0 q/2\hbar}$ cancels $e^{-\mathrm{i}qp_0/2\hbar}$, and Lemma 8.8 with $b = s$ and $k = -p_0/\hbar$ gives

$$\langle h_0, \varphi\rangle = \frac{e^{-q^2/4s^2}}{\sqrt{\pi}\,s}\int e^{-y^2/s^2} e^{\mathrm{i}p_0 y/\hbar}\,\mathrm{d}y = e^{-(q^2/s^2 + s^2p_0^2/\hbar^2)/4} = e^{-|\alpha|^2/2}.$$

   By Theorem 8.4, $\varphi = \sum_n \langle h_n, \varphi\rangle\, h_n = \psi_\alpha$.
2. The function φ is (4.3) with the width $s/\sqrt{2}$ and $c = e^{-\mathrm{i}qp_0/2\hbar}(\pi s^2)^{-1/4}$, and Theorem 4.9 gives the means and spreads.
3. The coefficient of h_n in the series is $e^{-|\alpha|^2/2}\alpha^n/\sqrt{n!}$, and the claim is its squared modulus. The mean of n is $\sum_n n e^{-|\alpha|^2}\,|\alpha|^{2n}/n! = |\alpha|^2$.
4. Since $e^{-\mathrm{i}\nu(n+\frac{1}{2})t}\alpha^n = e^{-\mathrm{i}\nu t/2}(\alpha e^{-\mathrm{i}\nu t})^n$ and $\left|\alpha e^{-\mathrm{i}\nu t}\right| = |\alpha|$, the series for $U_t\psi_\alpha$ is $e^{-\mathrm{i}\nu t/2}$ times the series for $\psi_{\alpha e^{-\mathrm{i}\nu t}}$. ∎

By the uniqueness in Theorem 5.4, the spectral measure of H is the one built from the h_n in Proposition 5.5. So, by Postulate 5.6, an energy reading on ψ_α gives $\hbar\nu(n+\frac{1}{2})$ with the probability $e^{-|\alpha|^2}\,|\alpha|^{2n}/n!$. Figure 8.3 draws these probabilities for $|\alpha|^2 = \frac{1}{2}$ and $|\alpha|^2 = 3$. For $\alpha \neq 0$ every level has a positive probability, so an energy reading on ψ_α can give many values.

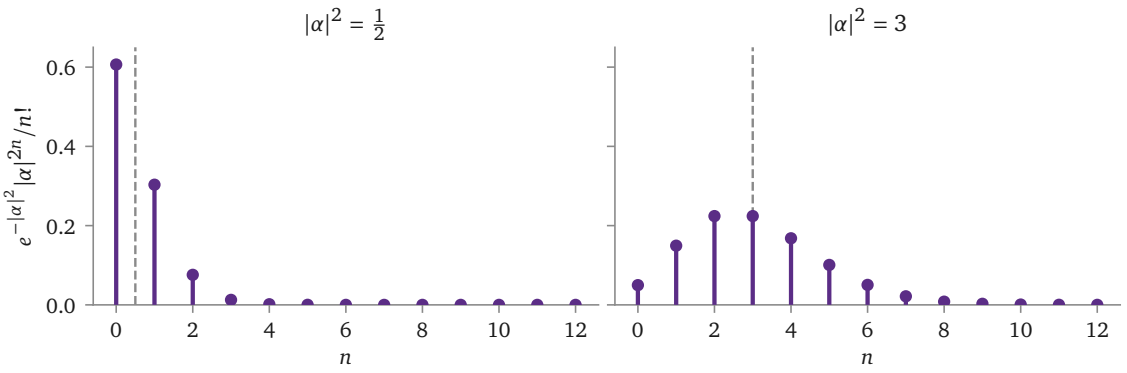


Figure 8.3: The probabilities $e^{-|\alpha|^2}\,|\alpha|^{2n}/n!$ of the level numbers $n = 0, \dots, 12$ in the coherent state ψ_α, for $|\alpha|^2 = \frac{1}{2}$ (left) and $|\alpha|^2 = 3$ (right), on one vertical scale. The stem at n is the probability that an energy reading gives $\hbar\nu(n + \frac{1}{2})$. The axis counts levels, not energy. The dashed line marks the mean $|\alpha|^2$ of n. The levels $n > 12$ are not drawn, and their total probability is below $2 \cdot 10^{-5}$ in both states.

Let q_t and p_t denote the means of position and momentum in $U_t\psi_\alpha$. The label $\alpha e^{-i\nu t}$ turns clockwise on the circle of radius $|\alpha|$. Written as $(q_t/s + is p_t/\hbar)/\sqrt{2}$, it has

$$q_t = q\cos\nu t + \frac{p_0}{m\nu}\sin\nu t, \qquad p_t = p_0\cos\nu t - m\nu q\sin\nu t,$$

which is the classical orbit from q and p_0, since $mq_t' = p_t$ and $p_t' = -m\nu^2 q_t$. By the theorem at $\alpha e^{-i\nu t}$, the mean position and momentum follow this orbit, and the spreads stay $s/\sqrt{2}$ and $\hbar/\sqrt{2}\,s$. A coherent state does not spread, unlike a free packet (Proposition 7.12). By the Poisson mean, its mean energy is $\hbar\nu(|\alpha|^2 + \frac{1}{2})$, and $\hbar\nu\,|\alpha|^2 = p_0^2/2m + m\nu^2 q^2/2$ is the classical energy of the orbit.

Figure 8.4 shows this motion for $q = s$ and $p_0 = 0$, where $q_t = s\cos\nu t$ and $p_t = -m\nu s\sin\nu t$. The packet moves and keeps its shape. The points of the circle are means, in one state at each time. They are not sharp values of position and momentum together.

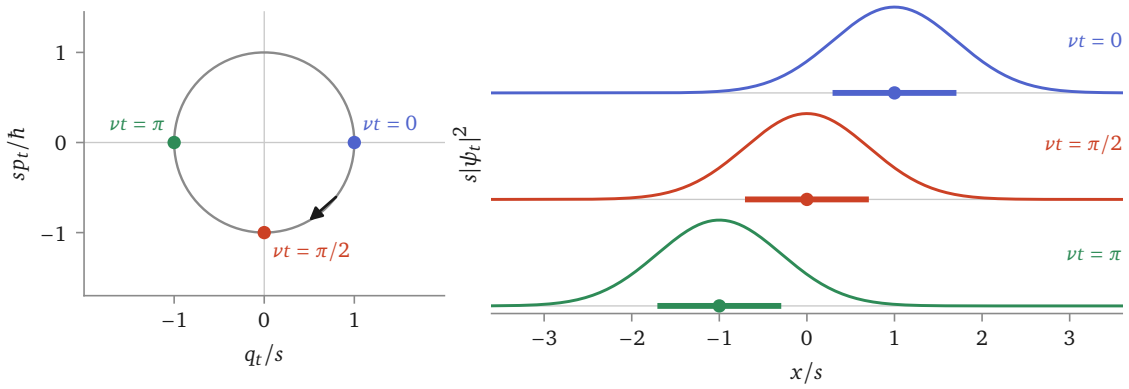


Figure 8.4: A coherent state with $q = s$ and $p_0 = 0$, at $\nu t = 0$, $\nu t = \pi/2$ and $\nu t = \pi$. The dots on the circle are the means of position and momentum, in scaled units. They turn clockwise on the unit circle. They are means, not a joint distribution of position and momentum. The curves are the position densities at the same times, one colour for each time, in one vertical scale, with the mean and one spread $s/\sqrt{2}$ on each side marked.

Self-check 8.5

1. Show that $\Delta Q\, \Delta P = \hbar/2$ in the state $U_t\psi_\alpha$, at every time t.
2. For $|\alpha|^2 = 2$, find the probability of the level number $n = 0$ and the mean energy.

8.6 Phase space inside the state space

The *phase space* of a classical particle on a line is the plane of its positions and momenta, and a point of this plane is a state of the classical particle. Each point (q, p_0) of this plane gives a state ψ_α, and different points give different means, so different states. The evolution maps these states to one another and turns them as the classical motion turns phase space. But the states overlap. The *overlap* of two states ψ_α and ψ_β is their inner product $\langle \psi_\alpha, \psi_\beta \rangle$. By the series and Parseval's identity,

$$\langle \psi_\alpha, \psi_\beta \rangle = e^{-(|\alpha|^2+|\beta|^2)/2} \sum_n \frac{(\overline{\alpha}\beta)^n}{n!} = e^{-(|\alpha|^2+|\beta|^2)/2+\overline{\alpha}\beta}, \qquad \left|\langle \psi_\alpha, \psi_\beta \rangle\right|^2 = e^{-|\alpha-\beta|^2}.$$

This is never 0. Two coherent states are never orthogonal, so no ideal reading tells them apart with certainty. Indeed, suppose a reading with the projection E answers yes with probability 1 on ψ_α and with probability 0 on ψ_β (Postulate 5.6). Since $\|\psi\|^2 = \|E\psi\|^2 + \|(\mathbb{1} - E)\psi\|^2$, this gives $E\psi_\alpha = \psi_\alpha$ and $E\psi_\beta = 0$. Since E is an orthogonal projection,

$$\langle \psi_\alpha, \psi_\beta \rangle = \langle E\psi_\alpha, \psi_\beta \rangle = \langle \psi_\alpha, E\psi_\beta \rangle = 0,$$

a contradiction. So a point of phase space is not an exact reading. Even the coherent states with real α, the translates of h_0, are enough to detect every nonzero vector. Lecture 9 uses this.

Proposition 8.11 (The translates of the ground state)**.** *If $\varphi \in L^2(\mathbb{R})$ is orthogonal to $h_0(x - q)$ for every real q, then $\varphi = 0$.*

Proof. For real $\alpha = q/\sqrt{2}\,s$, $h_0(x-q) = \psi_\alpha$ by Theorem 8.10. With $d_n = \langle h_n, \varphi\rangle$, the hypothesis reads $\sum_n d_n \alpha^n/\sqrt{n!} = 0$ for every real α. By Cauchy–Schwarz the power series converges absolutely, since $\sum_n |d_n|\,|\alpha|^n/\sqrt{n!} \le \|\varphi\|\, e^{\alpha^2/2}$. A power series that vanishes on $\mathbb{R}$ has only zero coefficients, since its derivatives at 0 vanish. So every $d_n = 0$, and $\varphi = 0$ by Theorem 8.4. ∎

Self-check 8.6

1. For which $|\alpha - \beta|$ is the squared overlap $\left|\langle \psi_\alpha, \psi_\beta\rangle\right|^2$ below 0.01? For which is the modulus $\left|\langle \psi_\alpha, \psi_\beta\rangle\right|$ of the overlap below 0.01?

The physics

In 1996 Meekhof, Monroe, King, Itano and Wineland held a single beryllium ion, $^9Be^+$, in a trap where its motion along one axis is an oscillator with $\nu/2\pi \approx 11.2\,\text{MHz}$. The oscillator length is then $s \approx 10\,\text{nm}$, and the ground state h_0 has $\Delta Q \approx 7.1\,\text{nm}$. Laser cooling left the ion in h_0 more than 95% of the time. A force oscillating at the trap frequency then prepared a coherent state. In the oscillator model such a force adds a term linear in Q to the Hamiltonian, and this evolution takes h_0 to a coherent state, up to a phase. The populations of the levels n, read through the evolution of the ion's internal levels, fitted a Poisson distribution: a fit of a coherent state gave $|\alpha|^2 \approx 3.1$, and a direct fit of the populations gave 2.9. For this α, $n = 0$ has the probability 0.045, the most likely value is $n = 3$, with 0.22, and the mean position swings with the amplitude $\sqrt{2}\,s\,|\alpha| \approx 25\,\text{nm}$.

The harmonic oscillator

States: the Hermite functions are an orthonormal basis of $L^2(\mathbb{R})$ (Theorem 8.4). Coherent states are Gaussian packets labelled by the points of phase space, and two of them are never orthogonal. *Observables:* the oscillator's energy is the only self-adjoint operator that acts on $\mathcal{S}(\mathbb{R})$ by $-(\hbar^2/2m)\psi'' + (m\nu^2/2)x^2\psi$. Its levels $\hbar\nu(n+\frac{1}{2})$ are simple and form its whole spectrum, and its resolvent is compact (Theorem 8.7). *Evolution:* every state returns, up to the sign -1, after the period $2\pi/\nu$, and a quarter period gives a Fourier transform, up to a common phase and a change of scale (Proposition 8.9). A coherent state follows the classical orbit without spreading (Theorem 8.10). *Measurement:* in a coherent state, the level number n of an energy reading $\hbar\nu(n+\frac{1}{2})$ follows a Poisson distribution with the mean $|\alpha|^2$. Two coherent states cannot be told apart with certainty.

Summary.

- *Ladder operators*: on $\mathcal{S}(\mathbb{R})$, $[a, a^\dagger] = 1$ and the oscillator's rule is $\hbar\nu(a^\dagger a + \frac{1}{2})$ (Proposition 8.2); $h_n = (n!)^{-1/2}(a^\dagger)^n h_0$ are orthonormal (Proposition 8.3).
- *The Hermite functions are complete* (Theorem 8.4).
- *The oscillator's levels*: the diagonal H is the only self-adjoint operator that acts by the oscillator's rule on $\mathcal{S}(\mathbb{R})$; $\sigma(H) = \{\hbar\nu(n+\frac{1}{2})\}$, simple and discrete, with a compact resolvent (Proposition 8.5 and Theorem 8.7).
- *A quarter period*: the Gaussian integral of Example 4.12 with $s = b/2$ is Lemma 8.8; $Fh_n = (-\mathrm{i})^n Dh_n$ and $U_{\pi/2\nu} = e^{-\mathrm{i}\pi/4} D^{-1} F$ (Proposition 8.9).
- *Coherent states*: Gaussian packets with $a\psi_\alpha = \alpha\psi_\alpha$, Poisson level numbers and the classical

orbit (Theorem 8.10); the closed span of the translates of h_0 is $L^2(\mathbb{R})$ (Proposition 8.11). Lecture 9 uses h_0 and its translates to prove the Stone–von Neumann theorem: under the conditions stated there, the canonical pair is unique up to unitary equivalence.

Exercises. Exercise Sheet 8.

Self-check answers.

8.1 1. $h_2 = a^\dagger h_1/\sqrt{2} = (2x^2/s^2 - 1)h_0/\sqrt{2}$, and $\langle h_0, h_2\rangle = (2\langle x^2\rangle_{h_0}/s^2 - 1)/\sqrt{2} = 0$, since the variance of h_0 is $s^2/2$.

8.2 1. Write $\varphi = \varphi_+ + \varphi_-$, with $\varphi_\pm(x) = (\varphi(x) \pm \varphi(-x))/2$. Since h_n has the parity of n, $\langle h_n, \varphi_+\rangle = 0$ for n odd, and $\langle h_n, \varphi_+\rangle = \langle h_n, \varphi\rangle = 0$ for n even. By Theorem 8.4, $\varphi_+ = 0$.

8.3 1. By Theorems 4.6 and 8.7, $\langle \psi, H\psi\rangle = \|P\psi\|^2/2m + m\nu^2\,\|Q\psi\|^2/2 \geq (\Delta P)^2/2m + m\nu^2(\Delta Q)^2/2$, and $u^2 + v^2 \geq 2uv$ with $u = \Delta P/\sqrt{2m}$ and $v = \sqrt{m/2}\,\nu\Delta Q$ gives at least $\nu\,\Delta Q\,\Delta P \geq \hbar\nu/2$.

8.4 1. Zero: by Proposition 8.9 the density is $m\nu\left|\tilde{\psi}(m\nu x)\right|^2$, which vanishes for $x < 0$.

8.5 1. $U_t\psi_\alpha$ is $\psi_{\alpha e^{-i\nu t}}$ up to a phase, which changes no spread, and by Theorem 8.10 every coherent state has $\Delta Q = s/\sqrt{2}$ and $\Delta P = \hbar/\sqrt{2}\,s$. 2. By Theorem 8.10, the probability of $n = 0$ is $e^{-2} \approx 0.135$, and the mean energy is $\hbar\nu(2 + \frac{1}{2}) = 5\hbar\nu/2$.

8.6 1. The squared overlap is $e^{-|\alpha-\beta|^2}$, and $e^{-|\alpha-\beta|^2} < 0.01$ exactly when $|\alpha - \beta|^2 > \ln 100 \approx 4.605$, that is, $|\alpha - \beta| > 2.146$. The overlap has the modulus $e^{-|\alpha-\beta|^2/2}$, and $e^{-|\alpha-\beta|^2/2} < 0.01$ exactly when $|\alpha - \beta|^2 > 2\ln 100 \approx 9.21$, that is, $|\alpha - \beta| > 3.035$. The squared overlap is the probability of yes, in the state ψ_β, for the projection onto the multiples of ψ_α.

References

Other books prove some results of this lecture. Hall, Thm 11.3, writes the Hermite functions with his polynomials $2^{-n/2}H_n$. Hall, Thm 11.4, proves Theorem 8.4 in the same way, with dominated convergence in place of the Taylor bound. Teschl, §8.3, proves the uniqueness claim of Theorem 8.7 with the smaller domain of the functions $r(x)e^{-x^2/2s^2}$, r a polynomial, in place of $\mathcal{S}(\mathbb{R})$, in his units, and in three dimensions in his Thm 8.5. Teschl, (8.46), states that the Hermite functions are eigenvectors of the Fourier transform.

Hall, *Quantum Theory for Mathematicians*, §§11.2–11.4 (pp. 228–236), Thm 11.3 (p. 233), Thm 11.4 (p. 235), Thm A.19 (p. 532). Teschl, *Mathematical Methods in Quantum Mechanics*, Thm 8.5 and (8.46) (p. 179). *Linear Algebra for Physics*, Ex. 13.35, Def. 14.19. Meekhof, Monroe, King, Itano and Wineland, *Phys. Rev. Lett.* 76, 1796 (1996).